

Supplementary Appendix for Edge4AV: Is Today's Edge Ready for Tomorrow's CAVs?

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A Additional Measurement Geography

The private 5G and V2X stationary collections used different locations and radio conditions. Figure 1 identifies the fixed infrastructure, radio samples, and stationary collection locations without reporting exact coordinates. The V2X marker labeled R is the reference payload sweep. Markers 1–5 correspond to the five stationary points used in the signal-strength and packet-size analysis.

Point 1 lies close to the RSU in geographic distance, but a building intersects the direct path between the devices. This NLOS condition accounts for its low normalized V2X signal. The remaining points extend across LOS and weaker-signal conditions. Consequently, the stationary packet-delivery results are ordered by normalized signal strength rather than by distance from the RSU.

B Additional 5G Payload Results

Figure 2 isolates payload size within one complete stationary collection. Every plotted condition returned 1000 accepted responses. At 8 KiB, MQTT had p50/p95 RTTs of 46.1/57.7 ms and TCP had 82.2/109.8 ms. At 256 KiB, the corresponding values were 442.1/578.2 ms and 389.8/466.0 ms. At 1 MiB, both medians exceeded one second; at 2 MiB, MQTT reached 2135.9/2535.9 ms and TCP reached 3139.9/3632.1 ms. These measurements show the payload-size effect without combining locations or incomplete conditions. The horizontal lines are comparison thresholds, not deadlines assigned to named CAV functions.

Figure 3 retains variation across all five stationary payload sweeps. Each marker is a condition-level p50 RTT, and its vertical segment extends to p95. Open markers denote conditions with fewer than 1000 accepted responses. Compact 1 KiB transfers remain below the larger payload groups, but the cross-run range shows that payload size alone does not determine RTT. The weak-signal third run produced a partial 256 KiB MQTT condition with a p50 of 15.15 s, while the complete favorable conditions at the same size remained below one second at p50. At 1–2 MiB, the median p50 across runs is seconds-scale for both transports.

The two views answer different measurement questions. The complete sweep quantifies the payload-size trend under one condition, whereas the cross-run view exposes how radio and operating conditions change that trend. Neither figure reports detector computation time or completion time for a source object larger than the transferred payload.

C IPI Primitive Semantics

IPI is the application-level interface used to identify what a CAV exchange represents. It does not prescribe a radio, network protocol, perception model, or planner. Every envelope records a message identifier, send timestamp, intersection and source identifiers, communication path, and optional session and correlation identifiers. A response expiration time indicates freshness; the correlation identifier associates the response with its request.

The C++17 reference implementation represents this contract with `EnvelopeMetadata` and typed `Envelope<T>` objects. The public receiver and sender APIs expose V2X ingestion and retrieval, session registration and maintenance, service invocation and response retrieval, telemetry submission, broadcast requests, and cloud-export operations. The high-level wrapper generates a message identifier and send timestamp when they are absent and derives a request correlation identifier from `requestId` when necessary.

`StatelessUpdate` corresponds to an envelope carrying a J2735 message type, encoding, payload, and optional frame counter. Its freshness is determined from the envelope timestamp and application policy. `ServiceRequest` identifies the service, vehicle, request, and optional session and carries vehicle context and application data. Its `desiredHorizonMs` or `requestedHorizonMs` field is a planning horizon, not a network deadline. The 100, 500, and 1000 ms analysis budgets are applied to sender logs.

The canonical `IpiServiceRequest` encoding contains a one-byte service type, two-byte request identifier, one-byte option flags, an optional two-byte planning horizon, a two-byte data length, and the application bytes. The implementation validates a planning horizon of at most 10,000 ms and an application-data field of at most 65,535 bytes. The richer `CooperativeServiceMessage` carries a 16-byte session identifier, vehicle identifier, service class, response state, optional planning horizon, freshness, confidence, planning, perception, control, and offload data. Its validators bound waypoint, detected object, control command, and opaque payload counts before serialization.

`SessionRegister` carries a vehicle profile, role, requested services, and optional subscription or path preference; the returned descriptor supplies the session identifier, state, lease, heartbeat interval, and granted services. `Refresh` maintains the session through a heartbeat and optional telemetry. The reference implementation creates an ACTIVE session, keeps

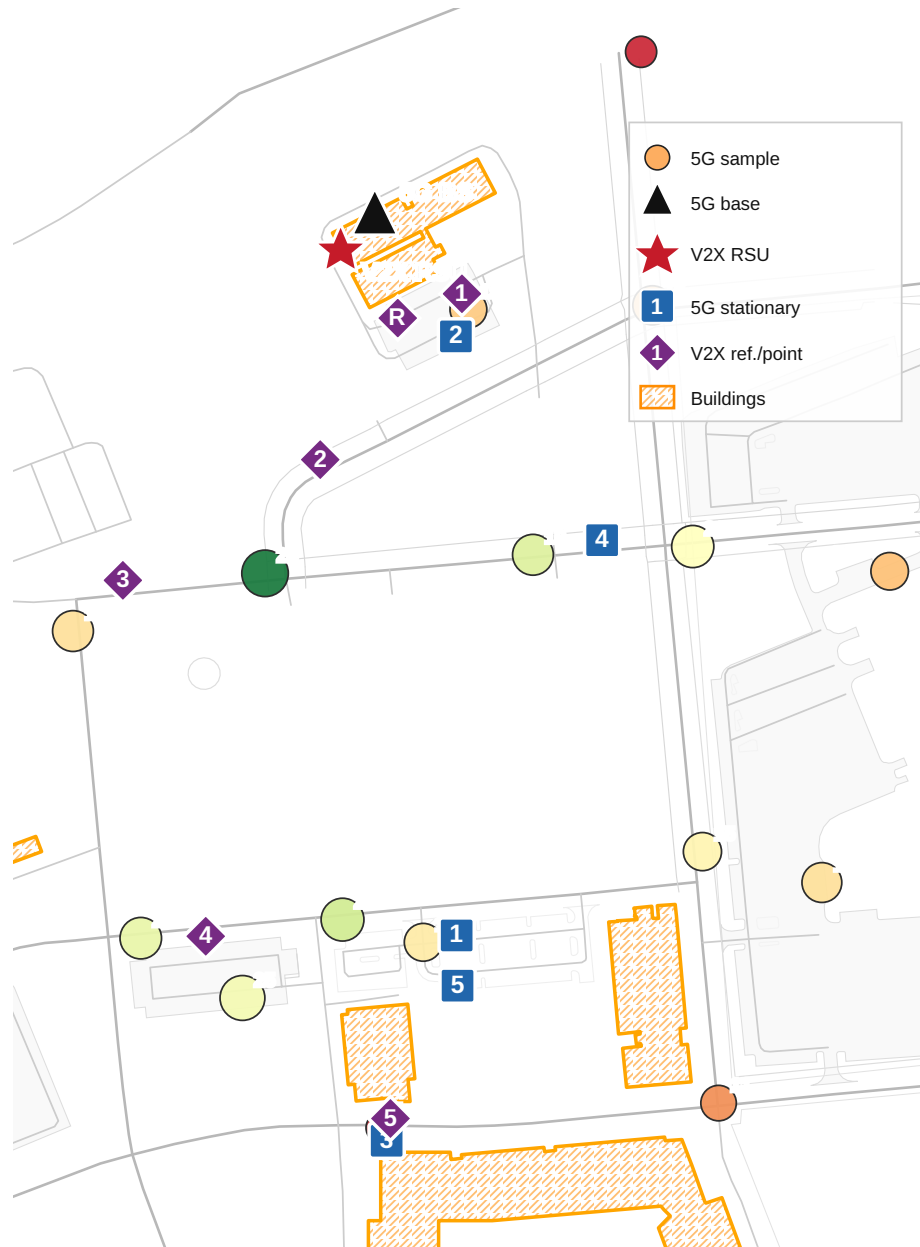


Figure 1: Stationary collection geography. Colored circles are 5G radio samples: fill color represents RSRP and the adjacent number reports SNR in dB. Blue squares and purple diamonds identify stationary 5G and V2X collections, respectively. The black triangle marks the 5G base station, the red star marks the V2X RSU, and orange hatching identifies building footprints.

it active on heartbeat or patch, and sets it to `TERMINATED` on termination. `SUSPENDED` is representable but is not exercised in the measurements. `ServiceResponse` carries accepted, rejected, in-progress, or completed status, guidance or result data, request correlation, and optional expiration.

An explicit `REJECTED` value is a remote response status. A local `Ack{accepted=false}` instead reports validation or API-admission failure and can include diagnostic text. The

current response type standardizes the rejection status but not a reject-reason enumeration. A local timeout, packet loss, or receiver/broker restart is an observed communication outcome and is not encoded as the same status. The logs preserve these outcomes separately: completed responses contribute RTT, explicit status remains part of the response record, and attempts with no response before timeout remain

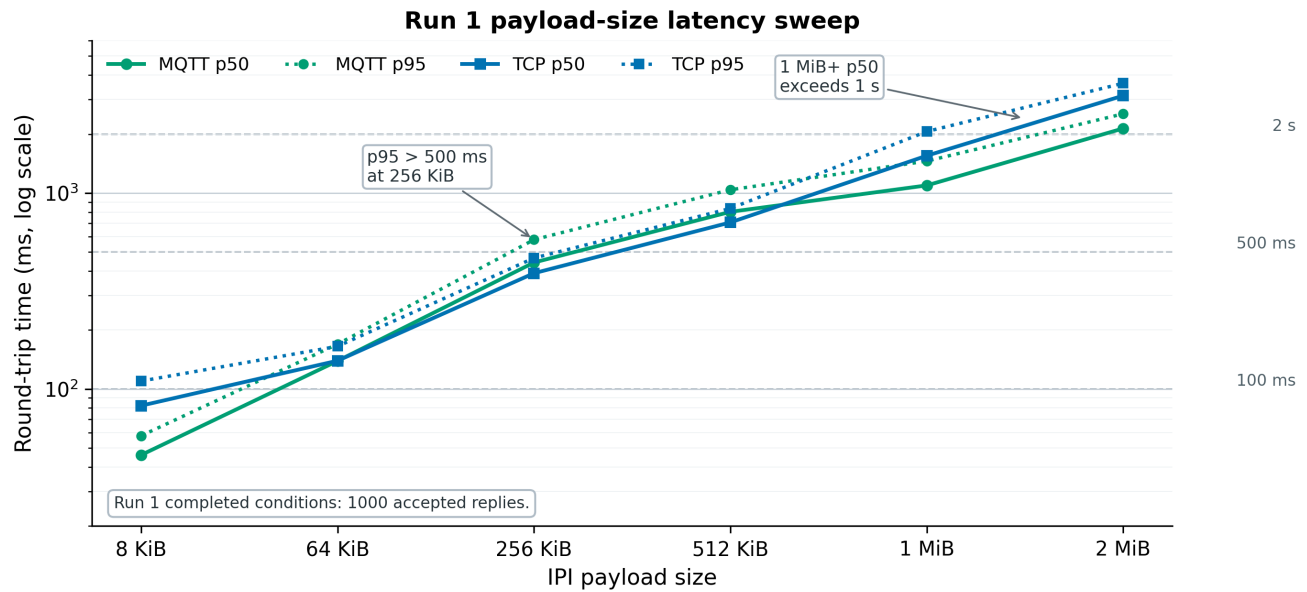


Figure 2: Complete stationary 5G payload sweep. Lines report p50 and p95 RTT for MQTT and TCP; all six payload conditions contain 1000 accepted responses per transport.

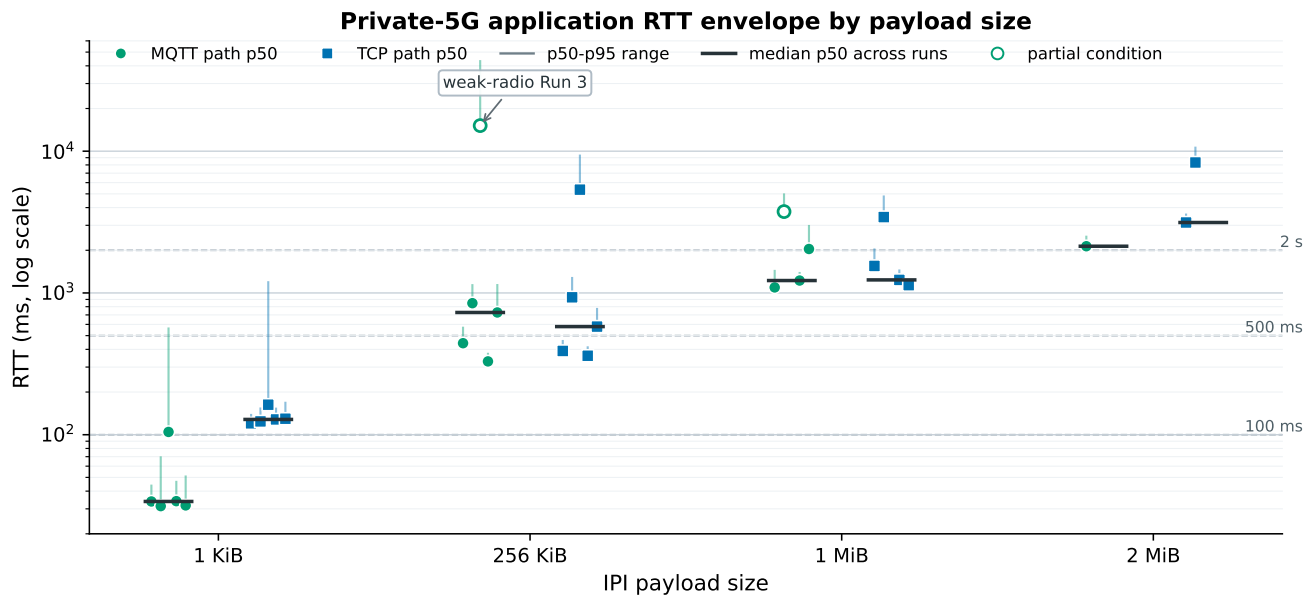


Figure 3: Stationary 5G payload results across five runs. Markers report p50 RTT and vertical segments span p50–p95. Filled markers contain 1000 accepted responses; open markers identify partial conditions. Horizontal lines are comparison thresholds.

in the response-availability and budget-miss denominators. Explicit service rejection is not benchmarked.

The V2X probes use request identity, sequence, payload size, timeout outcome, and RTT but do not time session registration or refresh. The 5G sender logs add session and service identifiers, protocol condition, acceptance/status, sender

Table 1: System boundary and operating parameters for the reported measurements.

Boundary	V2X radio path	5G network path	Interpretation
Endpoints	Vehicle-side Mocar OBU and fixed Mocar RSU	Vehicle-side sender through the Cisco radio/core network to a fixed edge-side host	RTT ends at the measured responder; cloud and Internet-backhaul delay are excluded.
Radio and service path	Certified production C-V2X devices; standard BSM/SPaT callbacks validated; vendor packet-data path used for correlated RTT probes	TCP receiver, MQTT 3.1.1 QoS 0 broker and receiver over TCP, or UDP receiver	The custom V2X probe supplies request correlation; it is not labeled as a standardized BSM or SPaT message.
Traffic generation	Stationary payloads 0–2048 B, 1000 attempts per condition, generally 100 ms interval	Payloads 0–2 MiB, generally 1000 attempts per condition, 200 ms interval, and 1–100 concurrent clients	Offered rate and concurrency determine application and network queueing.
Timeout and analysis budget	Stationary timeout 1000 ms; mobile timeout 500 or 1000 ms	UDP timeout 3–5 s in most runs and 8 s for the 60 KiB fragmented condition; MQTT timeout 60 s for detector replay and 30 s for multiclient runs	Timeout defines response availability. The 100/500/1000 ms budgets are analysis thresholds and can expire before the sender timeout.
Mobility and radio condition	Stationary LOS/NLOS locations and mobile 256 B probes at a 200 ms interval; median route speeds 7.13 and 7.54 m/s	Main payload, load, and multiclient runs are stationary; measured RSRP/SNR and weak-signal locations provide radio variation	V2X variation is interpreted using obstruction and route outcomes, not distance or motion alone.
Timing and device context	Sender-side RTT; the diagnostic interface did not provide usable received power	Sender-side RTT because endpoint clocks were generally unsynchronized	One-way delay is not used for claims. The V2X equipment is certified production hardware, and the cellular network is a Cisco deployment.

Table 2: Public V2X packet-size evidence and the deployed Mocar SDK limit.

Source	Interface or message	Reported size	Scope
Deployed Mocar SDK	Vendor packet-data interface	4,080 B maximum application packet	Static analysis of the SDK installed on the measured OBU and RSU; not a C-V2X standard limit.
Ettifos V2X-AIR manual [3]	C-V2X non-IP and IP interfaces	8188 B non-IP MTU; 1500 B IP MTU	Example of a different production device exposing implementation-specific packet limits.
CTI 4501 [1]	C-V2X transport	8000 B maximum including security overhead	Also recommends keeping SPaT and MAP messages small to maximize reception reliability.
FHWA CDA spectrum study [4]	BSM and SPaT	320 B typical; 1400 B maximum in the study model	Message scale used for safety and spectrum analysis, not a device MTU.

timestamps, and payload bytes. Dataset-derived replay uses the same request, payload, response, and timeout fields. Session registration and heartbeat structures are implemented, but their setup latency is outside the reported measurements.

D Measured Operating Regime

The V2X signal-strength map normalizes GNSS-joined latency and packet failures from the mobile runs. Failed attempts contribute the lowest value, while lower latency among successful attempts contributes a higher value. The stationary payload sweeps are separate experiments at locations on this map. Point 1 is NLOS because its direct path to the RSU intersects a building footprint.

The workload boundary contains 22,431 stored artifacts. Detector-output sizes are estimated from the measured number of predicted objects and are replayed directly over 5G. Encoded images, point clouds, and HD-map JSON files define larger source-object distributions. The 128–512 KiB and 1–2 MiB sweeps measure fixed transfer sizes or chunks rather than one-message encodings or completion times for every

full artifact. Detector processing time is reported separately and excluded from communication RTT.

These boundaries define the conditions under which the readiness limits were observed. The certified V2X devices and Cisco network avoid conclusions based only on prototype radios or emulated cellular links. The reported time-budget and response-availability values remain tied to the stated traffic, radio, and edge conditions.

E V2X Packet and Message Scale

The public Unex OBU-352 and Danlaw RouteLink Model 5 manuals describe C-V2X configuration and hardware but do not publish application-packet MTUs [2, 5]. No packet limit is inferred for those devices.

References

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